Comments from reviewer 1:

Recommendation: Reject

0. Lack of Conceptual and Practical Novelty

The manuscript presents the problem as being investigated "for the first time." However, the core components—bi-objective project selection, MDP modeling under uncertainty, and ADP/rollout algorithms for project scheduling—are well-established in the literature. The specific combination may be new, but the manuscript fails to clearly articulate the significant conceptual advance beyond existing work. The proposed algorithm, while competently designed, is essentially a sophisticated combination of known techniques (weighted-sum scalarization, Monte Carlo simulation, heuristic policies). A more rigorous discussion of the novelty relative to the rich body of work on ADP for project scheduling (e.g., Li & Womer, 2015; Satic et al., 2024, both cited in the manuscript) is critically needed.

1. Inadequate Benchmarking and Validation

The experimental design is the manuscript's most significant weakness and is insufficient to support the claims of superiority.

2.1 Unfair Comparison with Metaheuristic: The comparison with the multi-stage CQPSO is fundamentally flawed. By aggregating 20 dynamic stages into a single static stage and replacing uncertainties with their expected values, the authors create a simplistic and unrealistic benchmark. A proper comparison should be against other dynamic or online algorithms that can react to real-time information, not a crippled static approximation.

2.2 Lack of State-of-the-Art Baselines: The manuscript omits comparisons with more advanced and relevant benchmarks, such as other ADP variants, simulation-optimization techniques, or powerful multi-objective metaheuristics applied in a rolling-horizon framework.

2.3 Absence of Statistical Significance: The results presented in Tables 8 and 9 report average values but provide no measures of statistical significance (e.g., p-values from a Wilcoxon signed-rank test). It is impossible to determine if the observed performance differences are statistically meaningful or merely due to random variation.

2. Technical Omissions and Clarifications Needed

3.1 Normalization Function (Eq. 13): The normalization formula, Norm(x) = x / (max{x} - x), is highly problematic. The denominator max{x} - xcan be zero or negative, making the function undefined or nonsensical for objective values. This is a critical error that undermines the entire weighting mechanism in Equation (12). The standard approach is to use (x - min(x)) / (max(x) - min(x))or a similar range-based normalization.

3.2 Computational Efficiency Claim: The manuscript claims “computational efficiency” for the ADP algorithm. However, the reported CPU times (~13 seconds) are for instances with a planning horizon of T=100. Without reporting the time complexity or comparing runtime against other methods on a comparable basis (e.g., time to achieve a certain solution quality), this claim is unsubstantiated. The reliance

on M=100 simulations and Nω=7 weight samples per decision point suggests the algorithm is computationally intensive for real-time decision-making in large-scale problems.
3.3 Pareto Set Size: The ADP algorithm produces very small Pareto sets (averaging 2.6-2.75 solutions), which is a limitation for decision-makers seeking a diverse set of alternatives. This should be discussed as a drawback.
3. Given the extent of these required revisions, my recommendation is to reject the manuscript.

Comments from reviewer 2:

1. Summary of contribution
The manuscript studies a dynamic project portfolio selection and scheduling problem under uncertainty, where the decision-maker must balance expected net present value (NPV) and expected delay costs over a finite planning horizon. The authors formulate the problem as a bi-objective Markov decision process, incorporating stochastic durations and cash flows, resource and precedence constraints, and then propose an approximate dynamic programming / rollout approach to derive implementable policies. The paper positions this contribution against the largely deterministic and static PPSP literature, arguing that their framework better captures the sequential, uncertain nature of real-world portfolio decisions. The numerical experiments illustrate how different policy settings affect the trade-off between NPV and delay, and the authors discuss managerial implications for project prioritization under limited resources.
2. Comments
2.1 Literature Review: The literature review is generally well-structured and reasonably up-to-date. The authors situate their work within project portfolio selection by distinguishing deterministic PPSP models, static formulations under uncertainty, and the smaller stream of multi-stage dynamic models. They also cite recent contributions (2020-2025) on NPV-maximizing and risk-aware formulations, and clearly motivate the remaining gap on jointly managing NPV and delay costs in a dynamic, uncertain setting via a bi-objective MDP/ADP approach. However, the review is heavily oriented toward optimization and metaheuristics. For a Humanities and Social Sciences readership, it would benefit from a broader engagement with the project portfolio management and governance literature (e.g., empirical and conceptual work on how managers balance financial value and schedule risk). In addition, the dynamic PPSP and ADP/RL strands could be expanded and synthesized more explicitly, with a clearer novelty comparison to recent bi-objective scheduling work such as Li et al. (2025)
2.2 Methodology:
The core methodological idea-formulating the bi-objective dynamic project portfolio selection problem as a finite-horizon MDP and solving it via an ADP/rollout scheme-is appropriate and well suited to the application. The two objectives (expected NPV and expected delay cost), uncertainty in durations and cash flows, and

resource/precedence constraints are all incorporated, but the exposition is often compressed and some modeling choices remain under-specified.
In particular, the MDP elements (state vector, action space, transition dynamics, reward functions) should be defined more explicitly in one place. At present, components of the state (e.g. project status, remaining duration, budget) are scattered across sections, and the connection between the stochastic assumptions and the induced transition probabilities is not fully transparent. Likewise, the bi-objective treatment appears to rely on a weighted-sum scalarization, but the exact form of the combined objective and the practical interpretation of the weight(s) for decision-makers are not clearly stated.
For clearer and more reproducible methodology, the authors should specify:
(i) the base heuristic policy used in the rollout,
(ii) the number of simulation trajectories and time steps per iteration,
(iii) any value-function approximation structure and its parameters (if used), and
(iv) stopping criteria and convergence checks.
A concise pseudo-code algorithm plus a table summarizing key model parameters (horizon, discount factor, cost/benefit distributions, resource capacities, etc.) would greatly improve clarity.

2.3 Empirical Analysis:

The empirical analysis is useful but could be strengthened in several ways.
i) I would encourage a more detailed sensitivity analysis: e.g., varying the NPV-delay weights, resource availability to show how robust the proposed policy is to modeling assumptions.
ii) it would help to report variability across simulation runs (e.g., confidence intervals for NPV and delay cost) rather than only mean values, and to provide more visualizations of the trade-off frontier between NPV and delays.
iii) if any real or industry-inspired data are available, even as a small case study, applying the method there would greatly enhance the practical credibility of the results.

3. Recommendation: Revision

The paper has a strong and potentially publishable methodological core, and the topic is relevant to project portfolio decision-making under uncertainty. However, to be suitable for a Humanities and Social Sciences outlet, it needs a clearer connection to the broader project management and governance literature and a richer empirical section that incorporates more systematic sensitivity analysis and comparisons to realistic managerial baselines, ideally including at least one real or calibrated case study. With these improvements, the manuscript could make a meaningful contribution to the journal's readership.

How to balance net present value and delay costs in dynamic project portfolio selection? A pareto-optimal policy under uncertainty

Abstract: In project management, the effective selection of project portfolios and efficient utilization of limited resources exert a direct impact on portfolio returns and delivery timelines, thereby rendering them crucial to project success. Furthermore, uncertainties are inherently inevitable in real-world projects, and this further complicates decisions associated with portfolio selection. Therefore, we propose and investigate the bi-objective dynamic project portfolio selection problem (BODPPSP), which aims to dynamically maximize the net present value (NPV) and minimize the total delay cost of the portfolio simultaneously under uncertainty. We also take into account the reinvestment of project revenues, alongside the exclusive and complementary interrelationships between projects and their effects on the cash flow of the portfolio. We formulate a bi-objective Markov decision process model for the BODPPSP. To efficiently solve the NP-Hard BODPPSP, we design a bi-objective approximate dynamic programming (ADP) algorithm, which generates Pareto-optimal policies by sampling from the weight space of the two objectives (NPV and delay costs) and incorporates a rollout policy. Based on large-scale simulation experiments, we evaluate the performance of our ADP algorithm and compare it with a state-of-the-art metaheuristic algorithm and five benchmark heuristic policies. The experimental results verify the effectiveness of our ADP algorithm in generating Pareto-optimal policies for balancing NPV and delay costs in dynamic project portfolio selection under uncertainty.

Keywords: project management; dynamic project portfolio selection; uncertainty; bi-objective optimization; approximate dynamic programming

1 Introduction

Annually, projects account for one fifth of global economic activity, producing up to 12 trillion dollars in economic value (Antonio Nieto-Rodriguez, 2021). Effective selection of a project portfolio is essential for firms seeking a sustainable competitive advantage. The project portfolio selection problem (PPSP) involves choosing a subset of candidate projects to form a portfolio so that the portfolio performance is maximized subject to financial, resource, and other constraints (Noro & Dias, 2023; Zhong et al., 2019). PPSP determines whether a firm pursues the right projects. Allocating resources across different project sets yields distinct technology and product portfolios and ultimately shapes a firm's market position and share (Bai et al., 2025; Li et al., 2024).

The PPSP is NP-hard, and its optimization models and solution algorithms have been extensively investigated (Hartmann & Briskorn, 2022; Kavadias, 2001; Li et al., 2023; Mohagheghi et al., 2019). Most existing studies assume a deterministic environment. Chu, Hsu, and Fehling (1996) are among the early researchers to explore this problem; they develop a decision support system for R&D portfolio selection. However, many scholars regard the work of Archer and Ghasemzadeh (Archer & Ghasemzadeh, 1998, 1999) as a turning point in the PPSP literature. Sefair et al. (2017) examine a mean-semi variance project portfolio selection problem. They formulate a mixed-integer quadratic programming model that maximizes expected return while controlling downside risk and propose a Benders decomposition algorithm to solve it. Urli and Terrien (2010) develop an integer programming model for corporate PPSP and introduce a metaheuristic decision approach that supports multi-objective optimization to obtain portfolios that jointly optimize economic, scientific, and social contributions.

Maximizing revenue and ensuring timely delivery are fundamental objectives in project portfolio management (Shafahi & Haghani, 2018; Vanhoucke et al., 2001). A project may generate substantial profits, yet still fail if sudden cash shortages arise, making the net present value (NPV) a critical benchmark for decision-makers. Shou and Huang (2010) formulate a 0-1 integer programming model that maximizes portfolio's NPV and propose an Iterative multi-unit combinatorial auction (MUCA). MUCA uses a distributed bidding mechanism to coordinate resource allocation and project selection effectively. Nikkhahnasab and Najafi (2013) develop two metaheuristic algorithms. Their simulated annealing implementation uses random swaps and probabilistic acceptance to escape local optima, while their genetic algorithm (GA) employs a dual-mode crossover and random-exchange mutation operators to enhance global search. Li et al. (2016) further incorporate project interruption and inter-project interactions into a nonlinear mixed-integer programming model. To facilitate solution, they tradeoff between interruptions and interactions and transform the model into three mixed-integer linear programming (MILP) formulations.

Delays in project completion can lead to financial losses and damage an enterprise's reputation.

Kao et al. (2006) investigate resource conflicts and schedule interruptions in project portfolio management under unexpected events and propose a three-stage decision framework triggered by events. In the first stage, high-level Petri nets generate conflict-free feasible schedules. In the second stage, activity-based costing estimates schedule costs via a two-level allocation from resources to activities and then to projects. In the third stage, the technique for order preference by similarity to ideal solution evaluates the trade-off between duration and cost to select the most suitable schedule. The framework is supported by algorithms for deadlock detection, schedule generation, cost allocation, and standardized weighted ranking, which enhance its applicability in dynamic contexts. Namazian (2025) studies project portfolio selection and scheduling and proposes a risk-driven bi-objective mathematical model. The model incorporates intra-project risks and inter-project dependencies. Bayesian networks quantify risk probabilities and cascading effects. A MILP model is formulated to maximize portfolio net profit and minimize total completion time. The augmented ε-constraint method is applied to solve the multi-objective problem. Similarly, Li et al. (Li et al., 2025) address the project portfolio selection and scheduling problem with a bi-objective model that maximizes NPV and minimizes total delay cost, incorporating project interactions and profit reinvestment. They propose an improved NSGA-III that enhances solution quality through customized encoding/decoding and neighborhood search.

The studies above assume deterministic settings. However, uncertainty pervades in real-world project management. For example, project delays often cause cash inflows to diverge from forecasts (Gutjahr & Froeschl, 2013). Portfolios derived from deterministic models therefore frequently fail to deliver their expected performance, which motivates growing interest in PPSP under uncertainty. From a modeling perspective, work on uncertain PPSP falls into two broad classes: single-stage static models and multi-stage dynamic models.

Single-stage static models treat the portfolio's lifecycle decision as a one-off choice made at the initial time based on available uncertainty information (for example, distributions of project returns or risk probabilities). These models are simple and computationally efficient but cannot adapt to uncertainty that unfolds over time, and thus are incapable of evaluating and controlling potential risks.(Acebes et al., 2024) Within the static framework, common methods for handling uncertainty include stochastic programming (Huang et al., 2016; Sefair & Medaglia, 2005), fuzzy programming (Ranjbar et al., 2022), and robust optimization (Dixit & Tiwari, 2020; Hesarsorkh et al., 2021). Huang et al. (2016) examine project selection and scheduling under complex interproject links and cash-flow uncertainty. They formulate a single-objective stochastic program that maximizes expected NPV subject to a variance-based risk constraint, apply a mean-substitution technique to address cash-flow volatility, and solve the model using a GA. Sefair and Medaglia (2005) tackle a similar problem. They model uncertain cash inflows and outflows, introduce binary

decision variables and linearization to render nonlinear risk terms tractable, and formulate a bi-objective stochastic program that maximizes expected NPV while minimizing its variance. Their solution approach combines Monte Carlo Simulation with a two-stage iterative algorithm. Zolfaghari and Mousavi (2021) broaden the scope to settings with dynamic resource usage, variable completion dates, and multi-mode activities. They represent uncertainty in durations and resource consumption with normalized triangular interval fuzzy numbers and propose a three-objective framework that balances maximizing remaining funds, minimizing total tardiness, and improving robustness. They transform the framework into a single-objective mixed-integer programming (MIP) formulation for solution.

Unlike single-stage static models, multi-stage dynamic models partition the portfolio lifecycle into a sequence of decision stages. At each stage decisions are revised based on realized outcomes from the previous stage and newly observed stochastic information, such as whether project risk events occurred or how market demand shifts. Multi-stage formulations capture intertemporal dependencies and enable adaptive resource allocation, but they are more complex and computationally intensive. As a result, studies that rigorously address multi-stage PPSP remain relatively limited. Fang et al. (2008) examine mixed-asset portfolio selection with multistage investment features. They propose a bi-objective mixed-integer stochastic programming model that maximizes terminal wealth and minimizes risk. Uncertainty is handled via scenario generation, and the model is reformulated into a single-objective MILP for solution. Salo et al. (2022) focus on canonical multi-stage decision problems including PPSP. Recognizing that uncertainty can be endogenous and that the no-forgetting assumption may not hold, they introduce a decision programming framework built on Influence Diagrams. The framework accommodates multiple objectives and diverse risk measures, and it reconstructs the original problem as an equivalent MILP so that off-the-shelf solvers can be applied efficiently.

In summary, most of the existing studies on project selection assume a deterministic environment, typically focusing on optimizing either NPV or project delivery independently (Vanhoucke et al., 2001; Zolfaghari & Mousavi, 2021). To the best of our knowledge, no previous study has jointly optimized NPV and project delays from a dynamic decision-making perspective in an uncertain environment. Therefore, we fill this gap by studying the bi-objective dynamic decision model and approximate dynamic programming (ADP) algorithm for the bi-objective dynamic project portfolio selection problem (BODPPSP). Specifically, our main contributions are as follows:

(a) We propose and study the BODPPSP, which simultaneously maximizes the NPV and minimizes the total delay cost of the portfolio under uncertainty. In the BODPPSP, we consider the following uncertainties: dynamic project arrival, project duration fluctuation, and uncertain project

revenue. Interrelationships among projects (i.e., substitution and complementarity) and revenue reinvestment are also taken into account. We formulate a bi-objective Markov decision process (MDP) model for the BODPPSP.

(b) To efficiently solve the NP-hard BODPPSP, we design a bi-objective ADP algorithm. This ADP algorithm samples from the weight space corresponding to the two objectives to generate Pareto-optimal policies. Additionally, it incorporates a heuristic-driven estimator for the expected cost-to-go functions.

(c) We perform extensive computational experiments on a benchmark dataset consisting of 240 instances. We determine the parameter values of our ADP using the Taguchi method for the design of experiment (DOE). Based on four performance measures, we compare our algorithm with an existing meta-heuristic algorithm and five independent heuristic strategies.

The rest of the paper is structured as follows. Section 2 formalizes the problem and illustrates the key components via examples. Section 3 presents the MDP formulation that integrates dynamic project arrivals, duration variability, and revenue uncertainty. Section 4 proposes a rollout-based bi-objective ADP algorithm. Section 5 reports numerical experiments that benchmark our method against the state-of-the-art metaheuristic and several heuristic strategies. Section 6 concludes and outlines future research directions.

2 Problem statement

Projects arrive sequentially within the planning horizon T. The project manager is required to select a subset of these projects and allocate resources to the selected ones. The planning horizon T is evenly divided into L stages. Without loss of generality, we assume that at most one project arrives in each stage l, where $l = 0,1,2,\cdots,L$. At any stage l, project i arrives with probability $p_{l,i}$, while the probability of no project arrival is $1-\sum_i p_{l,i}$. The arrival time of project i is denoted as t_i. The duration $\boldsymbol{d}_i$ of each project i is stochastic, following a random distribution with an expected value of $\bar{d}_i$. The start time of a selected project i is denoted by s_i, and h_i represents its planned duration (typically estimated by managers based on prior experience before decisions are made). The estimated completion time of project i is thus given by $s_i + h_i$. Due to various sources of uncertainty, this estimate may be violated, meaning that the actual completion time $s_i + \boldsymbol{d}_i$ may exceed the estimated completion time. The excess duration is defined as the project tardiness, expressed as $\phi_i = \max\{\boldsymbol{d}_i - h_i, 0\}$. For each unit of tardiness, project i incurs a unit tardiness cost ct_i. Once a project started, it cannot be interrupted.

During execution, each selected project consumes limited renewable resources (e.g., labor, equipment, tools). There are K types of renewable resources., where the availability of resource type k is denoted by R_k, for $k = 1, 2, \ldots, K$. For each project i, its demand for resource type k

is a fixed integer r_{ik}. If the aggregate resource consumption of all currently active projects is less than R_k, the remainder is idle. The amount of idle resource is given by $IR_k = R_k - \sum_{i \in ACT} r_{ik}$, where ACT denotes the set of currently active projects.

Each project i requires an upfront investment $c_{i,out}$ (i.e., a cash outflow) at its initiation. The firm holds initial capital C_0 at the start of the planning horizon. Because such outlays are typically controllable by the firm, we assume the cash outflow $c_{i,out}$ is deterministic. Upon completion, project i yields a return $\boldsymbol{c}_{i,in}$ (i.e., a cash inflow), which we model as stochastic. The realized inflow is only observed after project completion and therefore necessarily reflects the effects of project interactions; accordingly, we do not explicitly model inter-project interactions on cash inflows in this paper. In addition, a portion of the revenue realized upon project completion can be used to finance subsequent projects. The share of total revenue available for reinvestment is defined as γ.

Under uncertainty (in project arrival times, durations, and returns), and subject to renewable resource and financial constraints, the BODPPSP seeks to dynamically screen arriving projects to build a portfolio and allocate resources to selected projects so as to maximize the portfolio's expected NPV while minimizing total tardiness cost (TC). Solving the BODPPSP therefore requires obtaining a set of policies with Pareto-optimal characteristics (covering both project selection and execution ordering) to support managerial decision-making.

Next, we provide an example to illustrate the BODPPSP. In this example, the planning horizon is defined as $T = 4$. The example involves a single type of renewable resource ($K = 1$), with a constant availability of $R = 6$. At the beginning of the horizon, the firm possesses initial capital of $C_0 = 20$. The entire project revenue is assumed to be reinvestable, corresponding to a reinvestment coefficient of $\gamma = 1$. Furthermore, the discount factor applied to cash flows is set to $e^{-\alpha} = 0.9$. Detailed information on the arriving projects is reported in Table 1.

Table 1 Project Information

Project No. (i)	$\boldsymbol{t_i}$	p_i	r_{ik}	$c_{i,out}$	$\bar{c}_{i,in}$	h_i	ct_i
1	0	0.7	4	10	21	1	2
2	1	0.7	2	8	28	3	4

Under an uncertainty environment, variations in project arrival sequences and in project start-time decisions can lead to markedly different resource allocations and timings of cash flows, which in turn substantially affect the objective measures such as the portfolio's NPV, total TC, and completion times. Figure 1 depicts a set of feasible solutions.

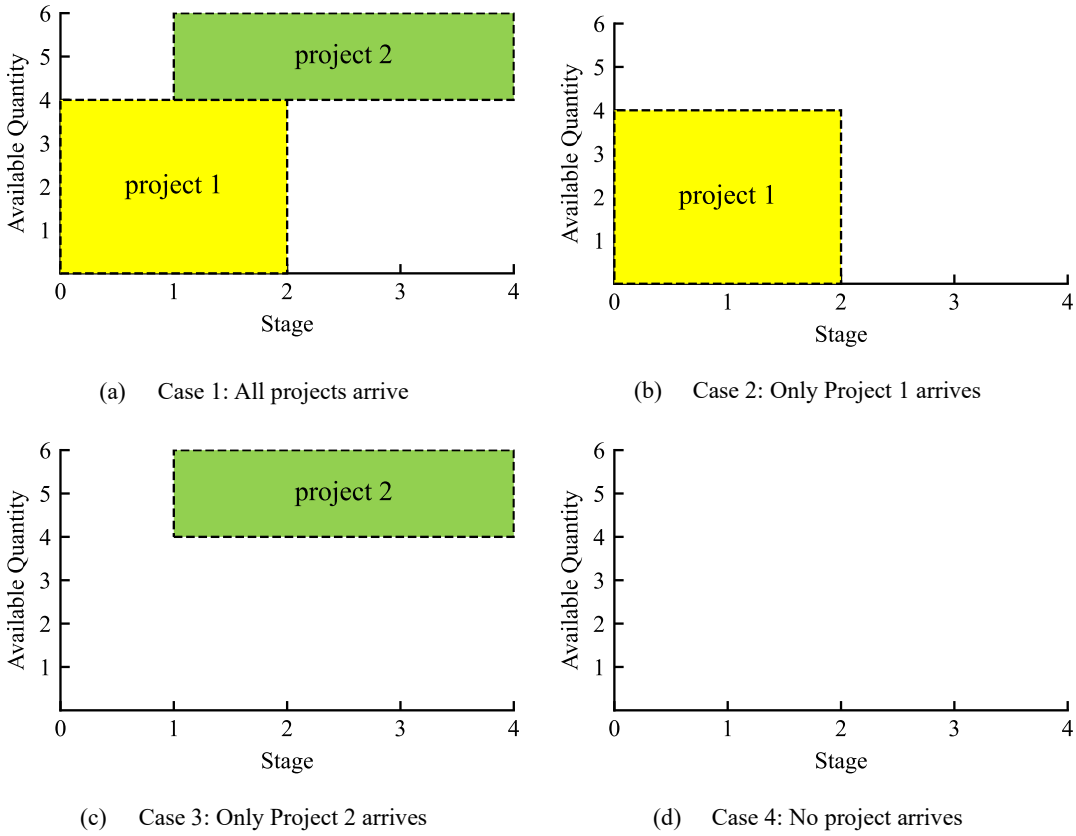


Figure 1 The Decision Results

For the example above there are four arrival scenarios; taking Case 1 as an example: At time 0 project 1 arrives. Project 1 is selected and initiated at stage 1. The remaining capital becomes $C_1 = C_0 - 10 = 10$, and the available resource vector is $\mathbf{R}'_1 = [2]$. At the end of stage 1 project 1 has not finished on schedule. Project 2 then arrives, so it is selected and scheduled to start immediately. By the end of stage 2 project 1 completes; its realized duration is 2 and its realized cash inflow is 18. Consequently, the remaining capital is $C_2 = C_1 - 8 + 18 = 20$, and the available resource vector is $\mathbf{R}'_2 = [4]$. At the end of stage 2 no project completes and no new project arrives, hence the remaining capital and available resources remain unchanged: $C_3 = C_2$ and $\mathbf{R}'_3 = \mathbf{R}'_2$. Project 2 completes on schedule at time 4, that is, at the end of stage 4. The realized cash inflow of project 2 is 30. Therefore the remaining capital becomes $C_4 = C_3 + 30 = 50$ and all resources are released so that $\mathbf{R}'_4 = [6]$.

Case 1 occurs with a probability of 0.49; under this scenario, the portfolio's NPV is 20.87, and the TC amounts to 2. Case 2 occurs with a probability of 0.21, yielding an NPV of 6.2 and a TC of 2 for the portfolio. Case 3 has a probability of 0.21, with the portfolio's NPV being 14.67 and TC being 2. Case 4 occurs with a probability of 0.09, where both the portfolio's NPV and TC are equal to 0. The expected portfolio's NPV is computed as the weighted sum, i.e., $\mathbb{E}[NPV] = 20.87 \times 0.49 + 6.2 \times 0.21 + 14.67 \times 0.21 + 0 \times 0.09 = 14.609$. Similarly, the expected total TC $\mathbb{E}[TC] = 1.82$.

3 Bi-objective Markov decision process model

In this section, we formulate the BODPPSP as a MDP model.

3.1 Stages and states

We partition the planning horizon T into L equally sized stages, indexed by $l = 0,1,2,\cdots,L$, where $l = 0$ denotes the initial decision epoch. The system state at the end of stage l is represented by $S_l = \{\mathbf{R}'_l, \boldsymbol{C}_l\}$ in which $\mathbf{R}'_l = [R'_{lk}, \forall k = 1,2,\cdots,K]$ denotes the vector of available renewable resource quantities at the end of stage l, and $\boldsymbol{C}_l$ denotes the available cash at that time. The state S_l therefore contains all information required to make decisions at stage l. The initial state is given by $S_0 = \{\mathbf{R}'_0, \boldsymbol{C}_0\}$, with $\mathbf{R}'_0 = [R_k, \forall k = 1,2,\cdots,K]$ and $\boldsymbol{C}_0 = C_0$. The terminal state is $S_L = \{\mathbf{R}'_L, \boldsymbol{C}_L\}$.

3.2 Decisions

At the end of each stage l, a decision must be made on whether to select the project that has newly arrived in that stage and to schedule it to begin at the start of the next stage. Specifically, at stage l we introduce a binary decision variable x_l, where $x_l = 1$ denotes selection of the newly arrived project i and $x_l = 0$ denotes rejection. If project i is selected, its start time is set to $s_i = t^{l-1}$, where t^{l-1} denotes the starting instant of stage $l-1$.

When making decision x_l, the required renewable resources and cash outflow of project i must not exceed the available amounts, i.e.,

$$x_l \cdot r_{ik} \leq R'_{l,k}, \qquad \forall k = 1,2,\cdots,K \tag{1}$$

$$x_l \cdot c_{i,out} \leq \boldsymbol{C}_l \tag{2}$$

If no project arrives in stage l, then $x_l = 0$.

3.3 State transition function

Let $T(\cdot)$ denote the state transition function that maps the system from the end of the current stage l to the end of the next stage $l+1$, evaluated before decisions are made at the latter stage. The next-stage state can therefore be written as $S_{l+1} = T(S_l, x_l, \omega_{l+1})$, where ω_{l+1} represents the realized stochastic information observed in stage $l+1$, that is, the information that becomes available after the implementation of decision x_l.(For example, ω_{l+1} may comprise the set of projects that complete during stage $l+1$, denoted C_{l+1}, together with the realized cash inflow $\boldsymbol{c}_{j,in}$ observed for each project $j \in C_{l+1}$.) It follows that the state S_{l+1} depends only on the current state S_l, the decision x_l, and the realized random information ω_{l+1}.

If i denotes the project that arrives in stage l, then the transition function $T(\cdot)$ is updated

with probability $p_{l,i}$ according to equations (3) and (4), and with probability $1-\sum_i p_{l,i}$ according to equations (5) and (6).

$$R'_{l+1,k} = R'_{l,k} + \sum_{j\in C_{l+1}} r_{jk} - x_l \cdot r_{ik}, \qquad \forall k = 1,2,\cdots,K \tag{3}$$

$$\boldsymbol{C}_{l+1} = \boldsymbol{C}_l + \gamma \cdot \sum_{j\in C_{l+1}} \boldsymbol{c}_{j,in} - x_l \cdot c_{i,out} \tag{4}$$

$$R'_{l+1,k} = R'_{l,k} + \sum_{j\in C_{l+1}} r_{jk}, \qquad \forall k = 1,2,\cdots,K \tag{5}$$

$$\boldsymbol{C}_{l+1} = \boldsymbol{C}_l + \gamma \cdot \sum_{j\in C_{l+1}} \boldsymbol{c}_{j,in} \tag{6}$$

Here, equation (3) gives the available resource at the end of stage $l+1$. It updates $R'_{l,k}$ by adding resources released by projects that complete in stage $l+1$ and subtracting resources consumed by projects that start in stage l. Equation (4) specifies the available cash at the end of stage $l+1$. It updates $\boldsymbol{C}_l$ by adding cash inflows from projects completed in stage $l+1$ and subtracting cash outflows from projects started in stage l. Equations (5) and (6) describe the analogous updates in the event that no new project is selected.

3.4 Cost functions

The single-stage cost function is the vector $\mathbf{f} = \langle f_{NPV}, f_{TC}\rangle$, whose components measure the incremental changes in the portfolio's NPV and total TC incurred by applying decision x_l in state S_l and transitioning to S_{l+1}.

$$\begin{aligned} f_{NPV}(S_l, x_l, S_{l+1}) \\ &= \sum_i p_{l,i}\left(\sum_{j\in C_{l+1}} \boldsymbol{c}_{j,in}\, e^{-\alpha\cdot t_j^c} - x_l c_{i,out} e^{-\alpha\cdot t^l}\right) \\ &+ \left(1-\sum_i p_{l,i}\right)\sum_{j\in C_{l+1}} \boldsymbol{c}_{j,in}\, e^{-\alpha\cdot t_j^c} \end{aligned} \tag{7}$$

$$f_{TC}(S_l, x_l, S_{l+1}) = \sum_{j\in C_{l+1}\cap DEL} ct_j\left(t^l - t_j^c\right) \tag{8}$$

where $e^{-\alpha} = (1+r)^{-1}$ denotes the discount factor that converts one monetary unit at the beginning of the next stage into its present value using discount rate r. (Leyman & Vanhoucke, 2017) The parameter ct_i denotes the unit TC of project i. The symbol C_{l+1} refers to the set of projects that complete in stage $l+1$, and DEL denotes the set of projects that are delayed. Equation (7) must account for two situations, whether a new project arrives in the stage or not. The change in NPV comprises cash inflows produced by completed projects and cash outflows caused by projects that start in the current stage. Equation (8) represents the sum of unit TC over all projects that are currently delayed.

3.5 Optimality equations

We seek a policy π that simultaneously maximizes the portfolio's NPV and minimizes its total TC. Let $\mathbf{J}_l(S_l) = \langle J_l^{NPV}(S_l), J_l^{TC}(S_l)\rangle$ denote the vector of expected cost-to-go functions starting from state S_l at stage l and extending to the terminal stage L. The terminal value is defined as $\mathbf{J}_L(S_L) = 0$. The corresponding optimality equations are given by:

$$J_l^{NPV}(S_l) = \max_{x_l}\{f_{NPV}(S_l, x_l, S_{l+1}) + \mathbb{E}[J_{l+1}^{NPV}(S_{l+1})]\} \tag{9}$$

$$J_l^{TC}(S_l) = \min_{x_l}\{f_{TC}(S_l, x_l, S_{l+1}) + \mathbb{E}[J_{l+1}^{TC}(S_{l+1})]\} \tag{10}$$

Solving the above equations yields a set of Pareto-optimal policies. A policy π is Pareto optimal if there exists no other policy π' such that for every objective $G \in \{NPV, TC\}$ one has $J_l^{G,\pi} \preccurlyeq J_l^{G,\pi'}$ and, furthermore, there is at least one objective G for which $J_l^{G,\pi} \prec J_l^{G,\pi'}$. When an objective is stated as a maximization problem (respectively a minimization problem), the relation $\preccurlyeq$ is equivalent to $\leq$ (respectively $\geq$) and $\prec$ is equivalent to $<$ (respectively $>$). The vector $\mathbf{J}_l(S_l)$ associated with a Pareto-optimal policy π is referred to as a Pareto-optimal point.

Equations (9) and (10) indicate that the optimal decision at stage l, denoted $\boldsymbol{x}_l(\pi^*)$, corresponds to a subset of the set of feasible activities X_l. Selecting the activities in $\boldsymbol{x}_l(\pi^*)$ minimizes the expected sum of the single-stage cost and the future expected cost starting at stage $l+1$. Therefore, by applying the recursion from $l = 0$ up to $L-1$, one can obtain optimal policies stage by stage in accordance with the dynamic programming (DP) principle. However, the BODPPSP is NP-hard and the presence of uncertainty causes the state space to grow exponentially with the number of projects. Exact DP thus suffers from the curse of dimensionality when computing $\mathbf{J}_l(S_l)$. To address this computational barrier, we develop an ADP algorithm tailored to efficiently approximate the solution of the optimality equation.

4 Approximate dynamic programming approach

We design an ADP algorithm based on the rollout approach. In the field of computer science, ADP are commonly referred to as reinforcement learning (Chu et al., 2019; Salemi Parizi et al., 2017; Zhao & Xu, 2021). ADP effectively alleviates the curse of dimensionality and offers scalable solution strategies for large-scale dynamic optimization problems. For example, many scholars have applied ADP to project scheduling problems under uncertainty (Satic et al., 2024; Xie et al., 2021). Choi et al. (2007) proposed an ADP method based on Q-Learning to address uncertainty in project arrival times. Li and Womer (2015) developed an ADP algorithm called ADP–HBA, that combines forward and backward reasoning to solve project scheduling problems with resource constraints and uncertain activity durations. Li et al. (2023) studied the resource leveling problem under triple

uncertainties in activity duration, overlap degree, and resource availability. They formulated a DP model and proposed a hybrid open-closed loop ADP (HOC-ADP) algorithm to efficiently solve this problem. Si et al. (2004) and Powell (2011) have provided comprehensive surveys for the ADP method.

A core design principle of ADP is to replace the future cost function $\mathbf{J}_{l+1}(S_{l+1})$ in equations (9) - (10) with an approximation. This approximation must satisfy two key properties: it should be computationally efficient and it should deliver high-fidelity estimates of the true future cost function (Li & Womer, 2015). Among the available techniques, rollout policies are particularly effective for constructing such approximations and are well suited to combinatorial optimization problems (Bertsekas et al., 1997). By using customized heuristic rules to estimate $\mathbf{J}_{l+1}(S_{l+1})$, the rollout approach can be seen as a heuristic extension of the policy-iteration method in dynamic programming (Bertsekas & Castanon, 1999).

4.1 ADP solution framework

To illustrate the basic rationale of our ADP, we first rewrite the optimality equation in (9) and (10) as:

$$\begin{aligned} J_l^G(S_l) &= \operatorname*{opt}_{x_l}\{f_G(S_l, x_l, S_{l+1}) + \mathbb{E}[J_{l+1}^G(S_{l+1})]\} \\ &= \operatorname*{opt}_{x_l}\left\{f_G(S_l, x_l, S_{l+1}) + \sum_{l_0=l+1}^{L-1} f_G(S_{l_0}, x_{l_0}, S_{l_0+1}) + \mathbb{E}[J_L^G(S_L)]\right\} \end{aligned} \tag{11}$$

where $G \in \{NPV, TC\}$. When the optimization objective is NPV, the operator opt denotes maximization, while for TC it denotes minimization.

In the following, we introduce two notations: the sample project portfolio's net present value $NPV(\sigma_{l+1}) = \sum_{l_0=l+1}^{L-1} f_{NPV}(S_{l_0}, x_{l_0}, S_{l_0+1})$, and the total tardiness cost $TC(\sigma_{l+1}) = \sum_{l_0=l+1}^{L-1} f_{TC}(S_{l_0}, x_{l_0}, S_{l_0+1})$. Here, σ_{l+1} denotes a state–decision trajectory from stage $l+1$, where project i arrives, to the end of the planning horizon (stage L).

To address different managerial risk preferences, a set of weight combinations is generated to construct the Pareto solution set. Monte Carlo simulation is employed to support the ADP exploration, enabling sequential computation of $NPV(\sigma_{l+1})$ and $TC(\sigma_{l+1})$ (see Section 4.2). In addition, to further enhance approximation quality, multiple heuristic strategies are used in parallel to guide the computational process (see Section 4.3). Based on these ideas, the overall ADP framework is as follows:

Step 0：Initialize the Pareto solution set $Pop = \emptyset$ Generate N_ω weight combinations from the normalized weight space $[0,1]$, forming the set Ω.

Step 1：For a specific weight combination (β_1, β_2), perform stage-by-stage selection. At each

stage l, estimate future costs using heuristic-driven expectation evaluation.。

Step 2：Determine the optimal decision x_l^*, update the set of active projects ACT, and record the state S together with the state-decision trajectory σ_0.

Step 3：If state S is terminal, compute the NPV and TC from the trajectory σ_0, and update the solution set as $Pop = Pop \cup \{(\sigma_0, NPC, TC)\}$

Step 4：Move to the next weight combination from Ω and return to Step 1. Once all weight combinations are processed, proceed to Step 5.

Step 5：Construct the Pareto frontier. The candidate solutions in Pop are sorted through Pareto dominance, yielding the final Pareto frontier $Pareto$.

4.2 Heuristic-based cost-to-go estimation

The core of cost-to-go estimation lies in assessing the expected future costs resulting from current decisions. Given the complexity and uncertainty inherent in multi-objective problems, it is infeasible to directly compute the expected future costs of all possible decisions at each stage. In practice, heuristic and metaheuristic algorithms have proven to be among the few effective approaches for large-scale, multi-stage problems. Therefore, we embed multiple heuristic strategies (see Section 4.3) into the cost-to-go estimation framework to further enhance the performance of ADP.

In each simulation, the heuristic strategies embedded in the estimation procedure are responsible for determining project portfolio selections in subsequent stages. For each feasible solution generated by these strategies, its performance is evaluated using Equation (12). Specifically, the portfolio's NPV (Equation (9)) and total TC (Equation (10)) are weighted according to a given weight combination (β_1, β_2). In this way, the variations in utility under different managerial risk preferences can be reflected for the same decision:

$$Obj(\sigma_l) = \beta_1 \cdot Norm\left(J_l^{NPV}(\sigma_l)\right) + \beta_2 \cdot Norm\left(J_l^{TC}(\sigma_l)\right) \tag{12}$$

Here, σ_l denotes a sample state-decision sequence obtained from Monte Carlo simulations. $J_l^{NPV}(\sigma_l)$ is equivalent to $J_l^{NPV}(S_l)$, and the same applies to $J_l^{TC}(\sigma_l)$.

Since the portfolio's NPV and total TC differ in both measurement units and value ranges, direct weighting would introduce bias. To eliminate dimensional inconsistencies and ensure comparability across different cost measures, normalized values are used in Equation (12), as defined in Equation (13). This ensures that the weight combination (β_1, β_2) accurately captures the contributions of NPV and TC to the overall objective.

$$Norm(x) = \frac{x}{\max\{x\} - x} \tag{13}$$

The pseudocode of the proposed heuristic-based cost-to-go estimation procedure is presented

in Algorithm 1.

Algorithm 1 **Heuristic-Based Cost-to-Go Estimation**

Input: Current state S_l, decision x_l, weight combination (β_1, β_2), set of ongoing projects ACT, number of simulations M, set of heuristic strategies HS

Output: Weighted estimation of future cost Obj_{x_l}

1 Initialize state–decision sequence σ_l

2 **FOR** $h \in HS$:

3 **FOR** $m = 1$ to M:

4   Execute decision x_l, update state S_l and sequence σ_l

5   **FOR** $l_0 = l + 1$ to T // Simulate subsequent stages

6   **IF** a new project j arrives **AND** its heuristic indicator falls within the acceptance domain

7   Set $x_{l_0} = 1$

8   **ELSE**

9   Set $x_{l_0} = 0$

10   Update ACT, state S and sequence σ_l

11   Compute the objective value Obj_m^h for sample m under strategy h

12 Compute average objective value $Obj_{avg}^h = \frac{1}{M} \cdot \sum_{m=1}^{M} Obj_m^h$

13 Compute $Obj_{x_l} = \max_{h \in HS}\{Obj_{avg}^h\}$

14 **RETURN** Obj_{x_l}

Algorithm 1 can be decomposed into three principal steps: applying the current decision, simulation and state generation, and approximation of the expected future cost.

For a decision x_l at stage l, Algorithm 1 first executes the current decision and observes the resulting state S_l. Based on this state, the algorithm generates M sample trajectories that cover stages $l + 1$ through L.

During the simulation phase (lines 5–10), each sample trajectory is simulated under $|HS|$ heuristic strategies, denoted by the set HS. For each future stage $l_0 \in \{l + 1, l + 2, \cdots, L\}$, if a newly arrived project j satisfies the resource and timing constraints and its heuristic indicator falls within the preconstructed acceptance domain (see Table 2), then the project is scheduled to start in the next stage by setting $x_{l_0} = 1$; otherwise, the project is ignored and $x_{l_0} = 0$. After each decision x_{l_0} is made, the state is updated using the state transition function. This procedure continues until all stages within the planning horizon have been simulated.

In the expected future cost approximation phase (lines 11–13), the state–decision sequences σ_l produced by the simulations are collected and used to compute Obj_{x_l} incrementally. The intention of this step is to evaluate feasible solutions from multiple perspectives under uncertainty, thereby improving the fidelity of the expected future cost estimates.

The heuristic-driven cost-to-go estimator balances computational efficiency and solution quality. By embedding this estimator into the ADP framework described in Section 4.1, the method is able to flexibly estimate the expected value of any decision at any stage.

4.3 Threshold tuning for the heuristic strategies

In the ADP framework, the effectiveness of a Rollout strategy is closely tied to how diverse and adaptable its baseline policies are. In uncertain settings, relying on a single heuristic often fails to capture the complexity and variability of real-world situations. To address this challenge, we design a rule base that supports multiple decision objectives and introduce a dynamic threshold mechanism, which together improve the robustness of the strategy.

Specifically, we integrate seven widely used heuristic rules into the cost-to-go estimator as a pool of candidate policies. The set includes return-focused rules (maximizing expected return, expected return rate, expected return rate per unit duration, and expected profit per unit duration), resource-oriented rules (minimizing total TC and expected duration), and a risk-averse rule (minimizing the probability of delay). Each rule addresses the project portfolio selection problem from a distinct perspective. In estimating future costs, we apply different combinations of these rules to generate a diverse set of solutions, which helps mitigate estimation errors in uncertain environments.

Most heuristics rely on scalar heuristic scores to drive selection. Accordingly, we design distinct heuristic indicators based on project attributes and construct an acceptance domain for each indicator. A candidate project i is selected when its indicator value falls within the corresponding acceptance domain; otherwise, the project is discarded. The specific indicators and their domain bounds are listed in the table below.

Table 2 Heuristic Indicators and Upper/Lower Bound Conditions

Strategy	Heuristic Indicator	Acceptance Domain	
		Lower Bound	Upper Bound
Minimum Possible Delay (PD)	$\frac{h_i - \bar{d}_i}{h_i}$	$-\infty$	Predefined Threshold
Minimum Unit Delay Cost (UDC)	ct_i		
Minimum Expected Duration (ED)	$\bar{d}_i$		
Maximum Expected Return (ER)	$\bar{c}_{i,in}$	Predefined Threshold	$+\infty$
Maximum Expected Return Rate (ERR)	$\frac{\bar{c}_{i,in}}{c_{i,out}}$		
Maximum Expected Return Rate per Unit Duration (ERRUD)	$\frac{\bar{c}_{i,in}}{c_{i,out} \times \bar{d}_i}$		
Maximum Expected Profit per Unit Duration (EPUD)	$\frac{\bar{c}_{i,in} - c_{i,out}}{\bar{d}_i}$		

The procedure for tuning optimal thresholds typically follows these steps. First, using small

batches of pre-generated data, we analyze the empirical distributions of each heuristic indicator, such as total TC and expected return rate, and we manually specify candidate threshold ranges. Second, we run each heuristic independently across the candidate thresholds and quantify the solution quality produced by each threshold setting. Single-policy optimal thresholds are selected according to hypervolume and the Epsilon indicator. For integrated use of multiple heuristics, orthogonal experimental design is employed to compare aggregate performance under multi-objective scenarios, and the best-performing threshold combination is chosen as the optimal set. By balancing local policy advantages and global synergy effects, this tuning process yields threshold parameters that are robust in dynamic environments.

5 Computational experiments

The proposed ADP algorithm was implemented in Python 3.7, and all computational experiments were carried out on a computer equipped with an Intel Core i7 2.30 GHz CPU running Windows 11 (64-bit). We begin by introducing the benchmark dataset and performance metrics, followed by a description of heuristic policy selection within the ADP framework and a discussion of optimal parameter settings for the algorithm. Next, we compare the proposed ADP method against static metaheuristics and several heuristic policies, and we analyze the comparative results in detail. Finally, we examine how variations in stochastic environment parameters affect the performance of the ADP algorithm.

5.1 Benchmark dataset

The benchmark dataset used in our experiments follows Kumar et al. (2018) We also account for uncertainty in actual project durations and inter-project interactions. The parameters used to generate the dataset are summarized in Table 3.

Table 3 Parameter Settings of the Benchmark Dataset

Parameter	Value
Environmental Parameters:	
Num. of resource types K	$U(1,5)$
Available quantity of resource R_k	$U(30,50)$
Initial capital C_0	$U(200,500)$
Planning horizon T	100
New project arrival probability p	0.7
Project Parameters:	
Unit cost of project delay ct_i	$U(1,10)$
Resource demand r_{ik}	$U(10,15)$
Mean duration $\bar{d}_i$	$U(5,20)$

Planned duration h_i	$U(0.8 * \bar{d}_i, 1.5 * \bar{d}_i)$
Actual duration $\boldsymbol{d}_i$	$\text{round}(d_{\min} + (d_{\max} - d_{\min}) \times \text{rand})$
Cash outflow $c_{i,out}$	$U(50,200)$
Planned cash inflow $\bar{c}_{i,in}$	$U(LB * c_{i,out}, UB * c_{i,out}) *$
Actual cash inflow $\boldsymbol{c}_{i,in}$	$\bar{c}_{i,in} * (Beta(2,5) * 0.7 + 0.8)$

In the benchmark instances we consider between one and five types of renewable resources. For each resource type k, the supply R_k is drawn from a uniform distribution over the interval $[30,50]$. To promote utilization of available resources, each project i is assigned a unit TC ct_i sampled uniformly from $[1,10]$.

The planning horizon is set to $T = 100$ and is divided evenly into stages. The mean duration of an arriving project, $\bar{d}_i$, is sampled from a specified uniform distribution. The planned duration h_i is an estimate of the actual duration $\boldsymbol{d}_i$ used to assess the likelihood of delay; h_i is drawn from a uniform distribution on $[0.8 * \bar{d}_i, 1.5 * \bar{d}_i]$. Following Li et al. (2023), the actual project duration $\boldsymbol{d}_i$ is generated by sampling from a right-skewed Beta distribution variant that has been widely used in the project selection literature (Deblaere et al., 2011; Van de Vonder et al., 2008). To obtain $\boldsymbol{d}_i$, for each project we first draw a random number rand from a $Beta$ distribution with parameters 2 and 5, and then compute $\boldsymbol{d}_i$ as follows:

$$\boldsymbol{d}_i = \text{round}(d_{\min} + (d_{\max} - d_{\min}) \times \text{rand}) \tag{14}$$

In Equation (14), $d_{\min} = 0.75 \times \bar{d}_i$ and $d_{\max} = 1.625 \times \bar{d}_i$. The symbol $\bar{d}_i$ denotes the mean duration of project i, and round $(\cdot)$ denotes rounding to two decimal places.

Cash-flow management plays a critical role in decision making. In our dataset the initial capital C_0 is drawn uniformly from $[200,500]$. The reinvestment coefficient is set to $\gamma = 0.8$ and the discount rate is $r = 0.01$. Project startup cash outflows $c_{i,out}$ follow a uniform distribution on $[50,200]$. The planned cash inflow $\bar{c}_{i,in}$ is defined as a random draw $U(LB * c_{i,out}, UB * c_{i,out})$, where LB and UB are lower and upper multipliers that reflect different return classes. Projects are categorized into low-, medium-, and high-return types according to the multipliers: low return has $LB = 1$ and $UB = 1.5$; medium return has $LB = 1.5$ and $UB = 2$; high return has $LB = 2$ and $UB = 2.5$. The realized cash inflow $\boldsymbol{c}_{i,in}$ is influenced by execution outcomes and is defined as the product of $\bar{c}_{i,in}$ and a fluctuation factor $Beta(2,5) * 0.7 + 0.8$, where $Beta(\cdot)$ denotes a Beta-distributed random variable.

We simulate three project arrival patterns. Scenario A represents a fully random arrival process, in which each arriving project is equally likely to be high-, medium-, or low-return with probability one third each. Scenario B models an environment where medium-return projects are predominant: medium-return projects arrive with probability 50% while high- and low-return projects each arrive with probability 25%. Scenario C represents an environment in which the firm actively screens for high-return applications: high-return projects arrive with probability 50%, and if no high-return

project is available the firm accepts medium- and low-return projects each with probability 25%. For each of the three scenarios we generate 80 instances, yielding a total of 240 benchmark instances.

5.2 Performance measures

We evaluate the performance of the ADP algorithm using the following five metrics:

(1) Size of the Pareto solution set ($Pareto$). This metric counts the number of non-dominated solutions produced by the algorithm. These solutions reflect different preference orientations and thus serve as important references for decision makers. A larger number of non-dominated solutions indicates a richer set of alternatives available to the decision maker; high-performing algorithms typically yield larger Pareto solution sets.

(2) Hypervolume (HV). The hypervolume metric quantifies the volume of objective space dominated by an approximate Pareto set relative to the entire objective space. (Zitzler et al., 2003) It jointly reflects the convergence and spread of the solution set, with larger values indicating better overall performance. Formally,

$$HV = \frac{\sum_{k=1}^{N} \sum_{i=1}^{C_N^k} (-1)^k V_i^k}{V_{total}} \tag{15}$$

where V_i^k denotes the volume of objective space dominated jointly by a specific combination of k solutions, N is the number of solutions in the approximate Pareto set, and V_{total} is the total volume of the objective space. To reduce redundant computation we adopt the inclusion–exclusion algorithm proposed by Tofighian and Naderi (2015) to compute the dominated volume of the entire Pareto set. When computing HV we set the objective-space bounds using the best (largest) NPV and the worst (largest) TC observed across all algorithm outputs. A larger HV indicates a stronger ability to obtain high-quality Pareto sets.

(3) Epsilon indicator (I_ε). The Epsilon indicator measures the minimal additive shift required to make an approximate Pareto set weakly dominate the optimal Pareto set (Zitzler et al., 2003). It is defined as:

$$I_\varepsilon = inf_{\varepsilon \in R} \{\forall\, z^2 \in B\ \exists\, z^1 \in A: z^1 \succcurlyeq_{\varepsilon +} z^2\}, \tag{16}$$

where A denotes the normalized optimal Pareto front (reference set), and B represents the normalized Pareto front obtained by a given algorithm. The relation $z^1 \succcurlyeq_{\varepsilon +} z^2$ holds if and only if $z_i^1 \leq z_i^2 + \varepsilon, \quad \forall i = 1, \dots, n$, meaning that solution z^1 ε-dominates solution z^2. In other words, after increasing each component of z^2 by ε, it will not be better than z^1. Equation (16) calculates the minimum increment ε required such that a solution in the approximate Pareto front is non-inferior to any solution in the optimal Pareto front. A smaller I_ε value indicates that the Pareto front obtained by the corresponding algorithm is closer to the optimal Pareto front.

(4) CPU time (CPU). This metric records the average runtime required by an algorithm to solve a single instance. Given comparable solution quality, shorter CPU time indicates higher computational efficiency.

5.3 Heuristic policy selection

In this section, seven commonly used heuristic strategies are compared, and a subset is selected as candidate heuristics for the ADP algorithm. Each heuristic strategy operates under the same principle: when the heuristic index value of a newly arrived project exceeds a given threshold, the project is selected and scheduled; otherwise, it is rejected.

Table 4 Performance of Different Heuristic Strategies

Strategy	Optimal Threshold	I_ε			
		Avg.	Gap	Max.	Min.
Min. PD	0.6	0.340	0.633	0.714	0.081
Min. UDC	5	0.346	0.415	0.597	0.182
Min. ED	12	0.428	0.425	0.656	0.231
Max. ER	250	0.453	0.575	0.751	0.175
Max. ERR	2	0.453	0.400	0.639	0.239
Max. ERRUD	0.1	0.415	0.560	0.717	0.157
Max. EPUD	10	0.549	0.531	0.783	0.253

The test dataset is constructed by randomly sampling 30 instances from each of the three categories of projects with different profit characteristics, resulting in a total of 90 instances. Based on this dataset, the performance of each heuristic strategy is reported in Table 4. According to the comparative results, four heuristics are ultimately selected: maximum expected return, maximum expected return rate, minimum expected project duration and maximum expected profit per unit of project duration.

5.4 Parameter settings

Proper parameter settings are crucial for enhancing decision-making performance. In this section, the test data set described in Section 5.3 and the orthogonal experimental design are employed to identify the optimal combination of parameters. Orthogonal experiments, as a systematic method for experimental design, are widely used in multi-factor, multi-level scenarios. They can effectively reduce the number of experiments while maximizing the exploration of relationships among factors, thereby optimizing the search process.

The ADP algorithm in this study involves two parameters: the number of Monte Carlo simulations (M) and the number of decision-preference samples (N_ω). Each parameter is tested at five levels (see Table 5). Accordingly, an $L_{25}(5^6)$ orthogonal array is selected, taking the first two columns of the array, which allows conducting 25 experiments across the two factors, each with five

levels.

Table 5 Combinations of parameter values

Factor level	Parameters (factors)	
	M	N_ω
1	20	3
2	50	4
3	100	5
4	150	6
5	200	7

The orthogonal array and the corresponding HV indicator values are presented in Table 6.

Table 6 Orthogonal array and hypervolume values

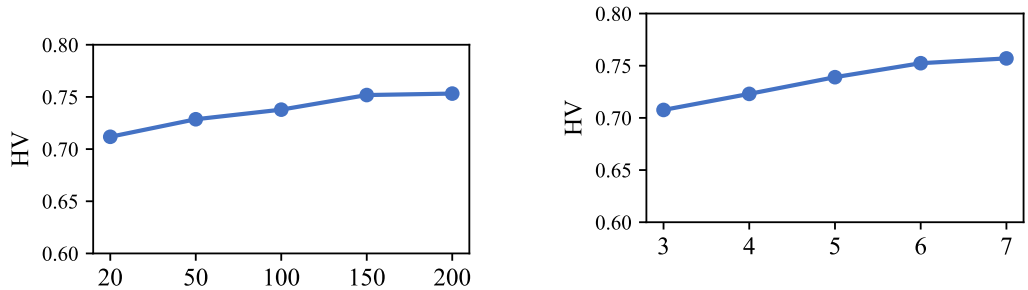
Experiment No.	Parameters (factors)		HV
	M	N_ω	
1	1	1	0.679
2	1	2	0.697
3	1	3	0.717
4	1	4	0.729
5	1	5	0.737
6	2	1	0.702
7	2	2	0.716
8	2	3	0.729
9	2	4	0.743
10	2	5	0.753
11	3	1	0.707
12	3	2	0.720
13	3	3	0.743
14	3	4	0.754
15	**3**	**5**	**0.765**
16	4	1	0.724
17	4	2	0.740
18	4	3	0.753
19	4	4	0.769
20	4	5	0.773
21	5	1	0.726
22	5	2	0.742
23	5	3	0.753
24	5	4	0.767
25	5	5	0.778

Based on these results, the hypervolume indicator is aggregated and classified to obtain the range and relative importance of each parameter, as summarized in Table 7.

Table 7 Response table for HV and rank for each factor

Level	M	N_ω
1	0.7118	0.7076
2	0.7286	0.723
3	0.7378	0.739
4	0.7518	0.7524
5	0.7532	0.757
Delta	0.0414	0.0494
Ranking	2	1

It is worth noting that for the ADP algorithm, higher parameter levels generally lead to longer computational times. Therefore, by jointly considering the hypervolume indicator and solution time, the final settings are determined as follows: the number of Monte Carlo simulations is set to $M = 100$ (level 3), and the number of weighted-space uniform samplings is set to $N_\omega = 7$ (level 5). Figure 2 provides a visual illustration of the impact of both parameters on solution performance.



(a) Num. of Monte Carlo simulations (M)

(b) Num. of weighted-space uniform samplings (N_ω)

Figure 2 Trends in parameter levels of the ADP

5.5 Main results

This section compares the proposed ADP algorithm with a metaheuristic algorithm and four high-performing standalone heuristic rules, as well as with a First Come First Served (FCFS) baseline.

(1) Comparison with metaheuristic algorithm

To facilitate a fair comparison, instances for the multi-stage cooperative quantum-behaved particle swarm optimization (CQPSO) method were constructed by aggregating every 20 original stages into a single new stage. Within each aggregated stage, uncertain project durations and returns were replaced by their expected values to form a deterministic integer programming model, which was then solved by the CQPSO algorithm. The objective values of the resulting project selection plans were computed using the actual realizations of the stochastic variables. The CQPSO

population size was set to 50 and the maximum number of iterations to 500.

Table 8 Comparison Results between the ADP and the Multi-stage CQPSO

Scenario	Algorithms	$Pareto$	HV	I_ε	CPU
A	ADP	2.750	0.731	0.263	13.058
	Multi-stage CQPSO	3.850	0.292	0.578	14.253
B	ADP	2.600	0.741	0.252	13.125
	Multi-stage CQPSO	3.925	0.204	0.596	13.028
C	ADP	2.750	0.763	0.232	12.587
	Multi-stage CQPSO	3.725	0.281	0.580	13.167

The results reported in Table 8 indicate that, although the multi-stage CQPSO as a static metaheuristic produces a relatively large Pareto set, the overall quality of that Pareto set is substantially inferior to the solutions obtained by the dynamic ADP approach. This outcome highlights an inherent disadvantage of static algorithms when applied to dynamic problems: even if a static method can find an optimal solution for a particular time window, its inability to access and exploit realized information in real time makes it difficult to attain global optimality across the full horizon. Under uncertainty, static metaheuristics struggle to reach good local optima within a short time frame.

(2) Comparison with heuristic strategies

In practical project management, especially under incomplete information and stochastic arrivals, online decisions are often driven by simple heuristic rules. When arrivals are sparse, decision makers may even adopt a FCFS heuristic to avoid resource waste. We therefore compare the proposed ADP method to the four superior heuristics selected in Section 5.3, as well as to the FCFS. The comparative results are summarized in Table 9.

Table 9 Comparison Results between the ADP and Heuristic Strategies

Scenario	Algorithms	$Pareto$	HV	I_ε
A	ADP	2.750	0.731	0.263
	Min. ED	1.0	0.443	0.435
	Max. ER	1.0	0.473	0.402
	Max. ERR	1.0	0.422	0.469
	Max. EPUD	1.0	0.575	0.313
	FCFS	1.0	0.309	0.545
B	ADP	2.600	0.741	0.252
	Min. ED	1.0	0.438	0.434
	Max. ER	1.0	0.471	0.394
	Max. ERR	1.0	0.480	0.396
	Max. EPUD	1.0	0.557	0.315
	FCFS	1.0	0.329	0.531
C	ADP	2.750	W0.763	0.232
	Min. ED	1.0	0.461	0.407

Max. ER	1.0	0.518	0.367
Max. ERR	1.0	0.442	0.421
Max. EPUD	1.0	0.576	0.316
FCFS	1.0	0.333	0.522

The results indicate that FCFS, while the computationally simplest, yields the poorest performance across all scenarios. The ADP algorithm, although slower than single heuristics due to its reliance on Monte Carlo simulation, produces multiple high-quality Pareto solutions that better support managerial choice. In both Hypervolume and Epsilon metrics, the approximate Pareto sets returned by the ADP method dominate those produced by the other heuristics.

Comparing ADP performance across scenarios reveals that the algorithm performs best under Scenario C, where high-return projects are more prevalent and the algorithm can more readily identify superior solutions. In Scenario B, where medium-return projects dominate and project returns are less differentiated, the algorithm has greater difficulty in distinguishing clearly superior choices.

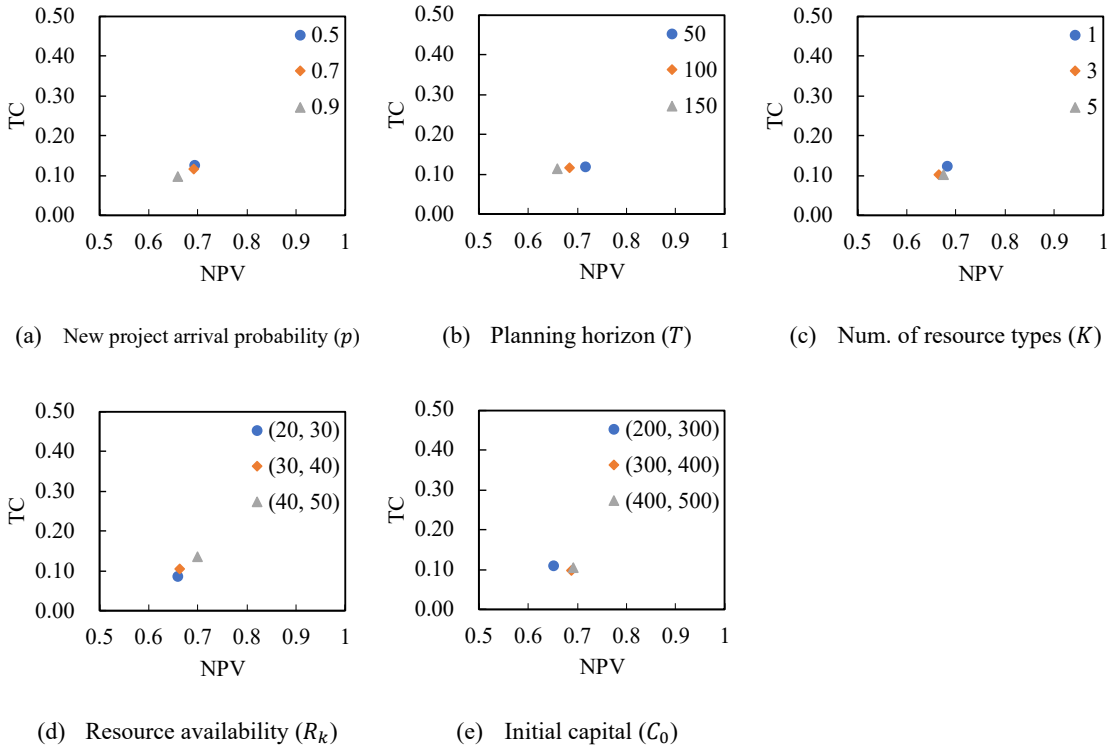
As the project environment becomes more complex and uncertainty increases, the dynamic capabilities of ADP are increasingly advantageous. Although ADP does not deliver instantaneous solutions like single-rule heuristics, its iterative simulation and online adaptation enable it to better accommodate uncertainty and thereby sustain a higher quality Pareto front. By contrast, static methods, lacking dynamic responsiveness, are prone to producing suboptimal or poor decisions in realistic project settings.

In summary, the proposed ADP algorithm not only generates a larger number of high-quality project-selection solutions for the BODPPSP, but its solutions are also closer to the theoretical Pareto frontier, demonstrating substantial practical value.

5.6 Sensitivity analysis

In practice, a dynamic interplay often exists among resource constraints, market fluctuations, and strategic objectives; organizations therefore need decision mechanisms with high flexibility. Although the proposed ADP algorithm demonstrates strong adaptability in multi-objective dynamic optimization, its practical value requires further validation under concrete managerial contexts.

This section conducts a systematic sensitivity analysis by varying key parameters in Scenario A, namely the new-project arrival probability p, planning horizon length T, number of resource types K, available quantity of resource R_k, and initial capital C_0. Figure 3 summarizes the results. Each marker in the figure represents the mean of the objective values within the approximate Pareto set obtained by the algorithm for a given parameter setting, after normalization. The horizontal axis denotes NPV and the vertical axis denotes total TC. Points located closer to the lower-right corner indicate better-quality solution sets.



(a) New project arrival probability (p) (b) Planning horizon (T) (c) Num. of resource types (K)

(d) Resource availability (R_k) (e) Initial capital (C_0)

Figure 3 Sensitivity Analysis Results

As the project arrival probability p increases, problem complexity rises and the algorithm's ability to find high-NPV solutions becomes more challenging. When $p \geq 0.7$, however, the project-selection space approaches saturation due to resource limits, and the rate of complexity growth attenuates; at the same time, the larger candidate pool facilitates achieving lower total TC. Extending the planning horizon T likewise increases overall problem complexity and can hinder the algorithm's capacity to secure high-NPV solutions. Because the per-unit-time complexity remains approximately unchanged, the impact of T on total TC is relatively minor.

The effect of the number of resource types K on problem difficulty exhibits no clear monotonic pattern. The influence of resource availability R_k is opposite to that of p: as resources become more abundant and constraints loosen, the algorithm more readily attains high-NPV solutions. Nevertheless, the concomitant increase in the solution space complexity can make it harder to locate solutions with low TC. Initial capital C_0, viewed as a financial resource, affects NPV in a manner similar to R_k, but its impact is attenuated by the reinvestment mechanism and is therefore weaker; its effect on total TC is also negligible.

Overall, across the tested parameter configurations, the algorithm's obtained solution sets exhibited an average normalized NPV of approximately 0.7 and an average normalized total TC of about 0.1. Apart from the planning horizon T, which exerts a noticeable influence on NPV, the remaining parameters only moderately affect algorithmic performance. Moreover, tuning algorithmic hyperparameters can mitigate many of the difficulties introduced by increased problem

complexity.

6 Conclusions and future work

In this study, the bi-objective dynamic project portfolio selection problem under uncertainty has been investigated for the first time. We formulate a bi-objective MDP to jointly optimize NPV and total TC. The model captures three sources of uncertainty: stochastic project arrivals, variability in project durations, and uncertain project returns. It also accounts for interrelationships among projects and enables the reinvestment of realized returns into subsequent projects. We develop a tailored ADP algorithm for the BODPPSP. This ADP algorithm is built on two principal components: sampling in the weight space and a heuristic-driven estimator for the expected cost-to-go. Furthermore, the ADP algorithm integrates multiple heuristic policies to enhance solution quality and strengthen robustness.

We conduct extensive large-scale computational experiments to validate the proposed ADP algorithm. The experimental findings demonstrate that the proposed ADP algorithm outperforms the selected metaheuristic methods and common heuristic approaches in terms of both solution quality and computational efficiency (i.e., computational time). Sensitivity analysis results indicate that higher project arrival rates and longer planning horizons tend to reduce the NPV, whereas greater resource availability and larger initial capital tend to increase the NPV.

For future research, investigating metaheuristic techniques to replace rule-based heuristics and enhance policy robustness would be of interest. Additionally, exploring exact methods for the BODPPSP represents a promising research direction.

References

Acebes, F., González-Varona, J. M., López-Paredes, A., & Pajares, J. (2024). Beyond probability-impact matrices in project risk management: A quantitative methodology for risk prioritisation. *Humanities and Social Sciences Communications*, *11*(1), 670. https://doi.org/10.1057/s41599-024-03180-5

Antonio Nieto-Rodriguez. (2021, November 1). The project economy has arrived. *Harvard Business Review*. https://hbr.org/2021/11/the-project-economy-has-arrived

Archer, N. P., & Ghasemzadeh, F. (1998). A decision support system for project portfolio selection. *International Journal of Technology Management*. https://doi.org/10.1504/IJTM.1998.014920

Archer, N. P., & Ghasemzadeh, F. (1999). An integrated framework for project portfolio selection. *International Journal of Project Management*, *17*(4), 207–216. https://doi.org/10.1016/S0263-7863(98)00032-5

Bai, L., Zhang, L., Zhang, L., Shao, K., & Luo, X. (2025). Unlocking the potential of project portfolio: Value-oriented interactive risk management. *Humanities and Social Sciences Communications*, *12*(1), 1012. https://doi.org/10.1057/s41599-025-05296-8

Bertsekas, D. P., & Castanon, D. A. (1999). Rollout Algorithms for Stochastic Scheduling Problems. *Journal of Heuristics*, *5*(1), 89–108. https://doi.org/10.1023/A:1009634810396

Bertsekas, D. P., Tsitsiklis, J. N., & Wu, C. (1997). Rollout Algorithms for Combinatorial Optimization. *Journal of Heuristics*, *3*(3), 245–262. https://doi.org/10.1023/A:1009635226865

Choi, J., Realff, M. J., & Lee, J. H. (2007). A Q-Learning-based method applied to stochastic resource constrained project scheduling with new project arrivals. *International Journal of Robust and Nonlinear Control*, *17*(13), 1214–1231. https://doi.org/10.1002/rnc.1164

Chu, P.-Y. V., Hsu, Y.-L., & Fehling, M. (1996). A decision support system for project portfolio selection. *Computers in Industry*, *32*(2), 141–149. https://doi.org/10.1016/S0166-3615(96)00067-X

Chu, Z., Xu, Z., & Li, H. (2019). New heuristics for the RCPSP with multiple overlapping modes. *Computers & Industrial Engineering*, *131*, 146–156. https://doi.org/10.1016/j.cie.2019.03.044

Deblaere, F., Demeulemeester, E., & Herroelen, W. (2011). Proactive policies for the stochastic resource-constrained project scheduling problem. *European Journal of Operational Research*, *214*(2), 308–316. https://doi.org/10.1016/j.ejor.2011.04.019

Dixit, V., & Tiwari, M. K. (2020). Project portfolio selection and scheduling optimization based on risk measure: A conditional value at risk approach. *Annals of Operations Research*, *285*(1–2), 9–33. https://doi.org/10.1007/s10479-019-03214-1

Fang, Y., Chen, L., & Fukushima, M. (2008). A mixed R&D projects and securities portfolio selection model. *European Journal of Operational Research*, *185*(2), 700–715. https://doi.org/10.1016/j.ejor.2007.01.002

Gutjahr, W. J., & Froeschl, K. A. (2013). Project portfolio selection under uncertainty with outsourcing opportunities. *Flexible Services and Manufacturing Journal*, *25*(1–2), 255–281. https://doi.org/10.1007/s10696-011-9107-2

Hartmann, S., & Briskorn, D. (2022). An updated survey of variants and extensions of the resource-constrained project scheduling problem. *European Journal of Operational Research*, *297*(1), 1–14. https://doi.org/10.1016/j.ejor.2021.05.004

Hesarsorkh, A. H., Ashayeri, J., & Naeini, A. B. (2021). Pharmaceutical R&D project portfolio selection and scheduling under uncertainty: A robust possibilistic optimization approach. *Computers & Industrial Engineering*, *155*, 107114. https://doi.org/10.1016/j.cie.2021.107114

Huang, X., Zhao, T., & Kudratova, S. (2016). Uncertain mean-variance and mean-semivariance models for optimal project selection and scheduling. *Knowledge-Based Systems*, *93*, 1–11. https://doi.org/10.1016/j.knosys.2015.10.030

Jennie Si, Andy Barto, Warren Powell, & Donald Wunsch. (2004). *Handbook of Learning and Approximate Dynamic Programming* (Vol. 2). John Wiley & Sons, Ltd. https://doi.org/10.1002/9780470544785.fmatter

Kao, H.-P., Wang, B., Dong, J., & Ku, K.-C. (2006). An event-driven approach with makespan/cost tradeoff analysis for project portfolio scheduling. *Computers in Industry*, *57*(5), 379–397. https://doi.org/10.1016/j.compind.2005.11.004

Kavadias, S. (2001). *Project portfolio selection and resource allocation in new product development* [English, INSEAD (France and Singapore)]. https://www.pqdtcn.com/thesisDetails/53556A44201CB24BA77102173EC8C702

Kumar, M., Mittal, M. L., Soni, G., & Joshi, D. (2018). A hybrid TLBO-TS algorithm for integrated selection and scheduling of projects. *Computers & Industrial Engineering*, *119*, 121–130. https://doi.org/10.1016/j.cie.2018.03.029

Leyman, P., & Vanhoucke, M. (2017). Capital- and resource-constrained project scheduling with net present value optimization. *European Journal of Operational Research*, *256*(3), 757–776. https://doi.org/10.1016/j.ejor.2016.07.019

Li, H., Hui, Q., & Wang, Y. (2024). Preemptive Software Project Scheduling Considering Personality Traits. *Systems*, *12*(10), 407. https://doi.org/10.3390/systems12100407

Li, H., & Womer, N. K. (2015). Solving stochastic resource-constrained project scheduling problems by closed-loop approximate dynamic programming. *European Journal of Operational Research*, *246*(1), 20–33. https://doi.org/10.1016/j.ejor.2015.04.015

Li, H., Yuan, Z., Yang, M., Yuan, Z., & Zhang, X. (2025). Project portfolio selection and scheduling considering net present value and delayed delivery. *Journal of the Operational Research Society*, *0*(0), 1–17. https://doi.org/10.1080/01605682.2025.2572996

Li, H., Zhang, X., Sun, J., & Dong, X. (2023). Dynamic resource levelling in projects under uncertainty. *International Journal of Production Research*, *61*(1), 198–218. https://doi.org/10.1080/00207543.2020.1788737

Li, X., Fang, S.-C., Guo, X., Deng, Z., & Qi, J. (2016). An extended model for project portfolio selection with project divisibility and interdependency. *Journal of Systems Science and Systems Engineering*, *25*(1), 119–138. https://doi.org/10.1007/s11518-015-5281-1

Mohagheghi, V., Mousavi, S. M., Antuchevičienė, J., & Mojtahedi, M. (2019). Project portfolio selection problems: A review of models, uncertainty approaches, solution techniques, and case studies. *Technological and Economic Development of Economy*, *25*(6), 1380–1412. https://doi.org/10.3846/tede.2019.11410

Namazian, A. (2025). A Novel Risk Driven Bi-objective Mathematical Model for the Problem of Project Portfolio Selection and Scheduling. *Journal of Applied Research on Industrial Engineering*, *Online First*. https://doi.org/10.22105/jarie.2025.462097.1615

Nikkhahnasab, M., & Najafi, A. A. (2013). *PROJECT PORTFOLIO SELECTION WITH THE MAXIMIZATION OF NET PRESENT VALUE*. *6*(12), 85–92.

Noro, J., & Dias, L. C. (2023). Project portfolio management considering the commitment of agents: A bi-objective model applied to administrative services. *Journal of the Operational Research Society*. https://www.tandfonline.com/doi/abs/10.1080/01605682.2022.2056530

Ranjbar, M., Nasiri, M. M., & Torabi, S. A. (2022). Multi-mode project portfolio selection and scheduling in a build-operate-transfer environment. *Expert Systems with Applications*, *189*, 116134. https://doi.org/10.1016/j.eswa.2021.116134

Salemi Parizi, M., Gocgun, Y., & Ghate, A. (2017). Approximate policy iteration for dynamic resource-constrained project scheduling. *Operations Research Letters*, *45*(5), 442–447. https://doi.org/10.1016/j.orl.2017.06.002

Salo, A., Andelmin, J., & Oliveira, F. (2022). Decision programming for mixed-integer multi-stage optimization under uncertainty. *European Journal of Operational Research*, *299*(2), 550–565. https://doi.org/10.1016/j.ejor.2021.12.013

Satic, U., Jacko, P., & Kirkbride, C. (2024). A simulation-based approximate dynamic programming approach to dynamic and stochastic resource-constrained multi-project scheduling problem. *European Journal of Operational Research*, *315*(2), 454–469. https://doi.org/10.1016/j.ejor.2023.10.046

Sefair, J. A., & Medaglia, A. L. (2005). Towards a model for selection and scheduling of risky projects. *2005 IEEE Design Symposium, Systems and Information Engineering*, 158–164. https://doi.org/10.1109/SIEDS.2005.193252

Sefair, J. A., Méndez, C. Y., Babat, O., Medaglia, A. L., & Zuluaga, L. F. (2017). Linear solution schemes for Mean-SemiVariance Project portfolio selection problems: An application in the oil and gas industry. *Omega*, *68*, 39–48. https://doi.org/10.1016/j.omega.2016.05.007

Shafahi, A., & Haghani, A. (2018). Project selection and scheduling for phase-able projects with interdependencies among phases. *Automation in Construction*, *93*, 47–62. https://doi.org/10.1016/j.autcon.2018.05.008

Shou, Y., & Huang, Y. (2010). Combinatorial auction algorithm for project portfolio selection and scheduling to maximize the net present value. *Journal of Zhejiang University SCIENCE C*, *11*(7), 562–574. https://doi.org/10.1631/jzus.C0910479

Tofighian, A. A., & Naderi, B. (2015). Modeling and solving the project selection and scheduling. *Computers & Industrial Engineering*, *83*, 30–38. https://doi.org/10.1016/j.cie.2015.01.012

Urli, B., & Terrien, F. (2010). Project portfolio selection model, a realistic approach. *International Transactions in Operational Research*, *17*(6), 809–826. https://doi.org/10.1111/j.1475-3995.2010.00762.x

Van de Vonder, S., Demeulemeester, E., & Herroelen, W. (2008). Proactive heuristic procedures for robust project scheduling: An experimental analysis. *European Journal of Operational Research*, *189*(3), 723–733. https://doi.org/10.1016/j.ejor.2006.10.061

Vanhoucke, M., Demeulemeester, E., & Herroelen, W. (2001). An Exact Procedure for the Resource-Constrained Weighted Earliness–Tardiness Project Scheduling Problem. *Annals of Operations Research*, *102*(1/4), 179–196. https://doi.org/10.1023/A:1010958200070

Warren B. Powell. (2011). *Approximate Dynamic Programming: Solving the Curses of Dimensionality* (1st ed.). Wiley. https://doi.org/10.1002/9781118029176

Xie, F., Li, H., & Xu, Z. (2021). An approximate dynamic programming approach to project scheduling with uncertain resource availabilities. *Applied Mathematical Modelling*, *97*, 226–243. https://doi.org/10.1016/j.apm.2021.03.048

Zhao, S., & Xu, Z. (2021). New closed-loop approximate dynamic programming for solving stochastic decentralized multi-project scheduling problem with resource transfers. *Expert Systems with Applications*, *185*, 115593. https://doi.org/10.1016/j.eswa.2021.115593

Zhong, Z., Li, X., Liu, X., & Lau, W. (2019). Opportunity cost management in project portfolio selection with divisibility. *Journal of the Operational Research Society*, *70*(7), 1164–1178. https://doi.org/10.1080/01605682.2018.1506546

Zitzler, E., Thiele, L., Laumanns, M., Fonseca, C. M., & Da Fonseca, V. G. (2003). Performance assessment of multiobjective optimizers: An analysis and review. *IEEE Transactions on Evolutionary Computation*, *7*(2), 117–132. https://doi.org/10.1109/TEVC.2003.810758

Zolfaghari, S., & Mousavi, S. M. (2021). A novel mathematical programming model for multi-mode project portfolio selection and scheduling with flexible resources and due dates under interval-valued fuzzy random uncertainty. *Expert Systems with Applications*, *182*, 115207. https://doi.org/10.1016/j.eswa.2021.115207

Funding

This research was supported by the Soft Science Research Project of the Shanghai Municipal High-Level Institution Construction and Operation Plan.

Author contributions

XX conceived the study and proposed the original idea. XX, XX, XX and XX designed the methodology and performed the data analysis. XX and XX wrote the first draft. XX and XX reviewed and revised the manuscript.

Competing interests

The author(s) declare no competing interests.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

No human subjects are involved in this study.

Data availability

The datasets generated and analyzed during the current study are not publicly available due to the fact that the key parameters and methods for data generation are described in detail in the main text, but are available from the corresponding author on reasonable request.